

Bulk-Type All-Solid-State Lithium Batteries Using Complex Hydrides Containing Cluster-Anions

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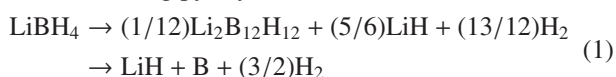
In this study, we incorporated fast lithium ion conducting complex hydrides containing cluster anions, namely icosahedral dodecahydro-closo-dodecaborate anions, $[B_{12}H_{12}]^{2-}$, into a bulk-type all-solid-state battery. $Li_2B_{12}H_{12}$, TiS_2 and Li were used as an electrolyte, a positive electrode active material and a negative electrolyte, respectively, for a battery assembly to investigate its battery performance. "Bulk-type" battery contains high quantity of electrode active materials in the electrode layer, and thereby the enhanced energy-storage density is expected. In addition, nonflammable solid-state electrolyte would ensure the safety, which is currently problematic for the conventional lithium rechargeable battery that uses organic liquid electrolyte. $Li_2B_{12}H_{12}$ has a high lithium ionic conductivity of $\log(\sigma/S\text{ cm}^{-1}) = -2.6$ at 393 K. It exhibits a relatively higher conductivity of $\log(\sigma/S\text{ cm}^{-1}) = -3.5$ at reduced temperatures such as 333 K. These high lithium ionic conductivity allows for the repeated operation of the bulk-type all-solid-state TiS_2/Li battery for at least 10 cycles at not only 393 K but also 333 K with the discharge capacities higher than 190 mAh g⁻¹. $Li_2B_{12}H_{12}$ exhibits higher oxidative stability. Thus, our battery realized high coulombic efficiencies of over 92% during the battery operation. This information will be beneficial for the further development of novel complex hydride-based electrolyte to have high-performance bulk-type all-solid-state batteries. [doi:10.2320/matertrans.MAW201601]

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1. Introduction

Complex hydrides possess energy-storage- and conversion-related functions including hydrogen storage,^{1,2)} lithium storage,^{3,4)} and fast ionic conduction.^{5–7)} Owing to their high hydrogen densities, we have devoted our significant research and development efforts to investigate complex hydrides as potential hydrogen storage materials. $LiBH_4$, a typical complex hydride, experiences a phase transition from an orthorhombic structure (low-temperature phase, LT phase) to a hexagonal one (high-temperature phase, HT phase) at elevated temperatures and around 390 K.⁸⁾ Following a further increase in temperature, this material releases hydrogen according to the following pyrolysis reaction:^{2,9–12)}



This pyrolysis enables the release of hydrogen from $LiBH_4$ with a gravimetric hydrogen density of 13.8 mass%. In the HT phase of $LiBH_4$, a nearly free rotation of the complex anion $[BH_4]^-$ occurs. This assists the formation of a metastable interstitial lithium site, leaving a lithium vacancy and enabling the material to act as a lithium carrier.^{13–15)} A percolation of lithium sites in its crystal lattice enables the formation of lithium ion conduction paths. These result in high lithium ionic conductivity that exceeds $\log(\sigma/S\text{ cm}^{-1}) = -3$ in the HT phase of $LiBH_4$.⁵⁾ Since our discovery of the fast lithium ionic conduction in the HT phase of $LiBH_4$, we explored complex hydrides that exhibit high ionic conductivities with the hope of incorporating them into the solid-state batteries. As a result, we found a variety of the complex hydride-based fast lithium and sodium ionic conductors with various crystal structures and compositions.^{6,7)}

Complex hydride electrolytes have numerous merits, par-

ticularly light weight, ability to accept metal negative electrodes, high-temperature durability, and simple handling, when they are incorporated into all-solid-state batteries.⁷⁾ Recently, we demonstrated the repeated operation of a bulk-type all-solid-state lithium rechargeable battery that uses the $LiBH_4$ -based electrolytes.^{7,16–20)} Hence, this class of the materials is considered to represent the third family of solid-state electrolyte, after sulfides and oxides. The solid-state battery that uses $LiBH_4$ could only operate at temperatures above 393 K, at which point, the $LiBH_4$ electrolyte exhibited high lithium-ionic conductivity. Thus, a novel complex hydride electrolyte that decreases the battery operating temperature is urgently required.

The successful repeated operation of solid-state batteries relies on not only realizing high lithium ionic conductivity but also introducing a robust interface that hinders the mutual diffusion of constituent elements across the electrode and electrolyte since the mutual diffusion leads to capacity fading,^{21,22)} and the hetero-interface comprised of the electrode and electrolyte often invites an increased interface resistance owing to the formation of a space charge layer.^{23–25)}

We have previously investigated the factors influencing the stable operation of the bulk-type all-solid-state TiS_2/Li batteries using the $LiBH_4$ electrolyte, paying careful attention to the interface stability between the positive electrode active material and the solid state electrolyte in the positive electrode layer. The precipitates accompanied by hydrogen desorption from the interface region, most likely $Li_2B_{12}H_{12}$,^{2,9–12)} act as a stable interface that has high electrochemical stability and charge transfer reactivity. As a result, over 300 discharge-charge cycles were successfully demonstrated.¹⁸⁾ We explored a series of complex hydrides having a cluster anion as a novel electrolyte design guide, and discovered various fast lithium^{26,27)} and sodium ionic conductors;^{26–30)} the ionic conductivities of a selection of these compounds are shown in Fig. 1.

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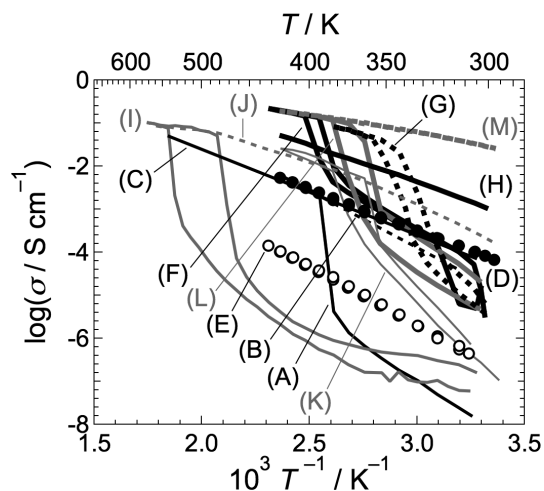


Fig. 1 Lithium ionic conductivities of (A) LiBH_4 ,⁵⁾ (B) $\text{Li}_4(\text{BH}_4)_3\text{I}$,³¹⁾ (C) $\text{Li}_2\text{B}_{12}\text{H}_{12}$,³²⁾ (D) $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$, (E) $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$, (F) $\text{LiCB}_{11}\text{H}_{12}$,²⁶⁾ (G) $\text{LiCB}_9\text{H}_{10}$,²⁷⁾ and (H) $90\text{LiBH}_4:\text{P}_2\text{S}_5$.¹⁶⁾ Sodium ionic conductivities of (I) $\text{Na}_2\text{B}_{12}\text{H}_{12}$,²⁸⁾ (J) ball-milled $\text{Na}_2\text{B}_{12}\text{H}_{12}$,²⁹⁾ (K) $\text{Na}_2\text{B}_{10}\text{H}_{10}$,³⁰⁾ (L) $\text{NaCB}_{11}\text{H}_{12}$,²⁶⁾ and (M) $\text{NaCB}_9\text{H}_{10}$.²⁷⁾

In this study, we have incorporated the $\text{Li}_2\text{B}_{12}\text{H}_{12}$ electrolyte, which was recently shown by He and coworkers to exhibit high lithium ionic conductivity at temperatures lower than 393 K (for instance, $\log(\sigma/\text{S cm}^{-1}) = -3.6$ for $\text{Li}_2\text{B}_{12}\text{H}_{12}$ ³²⁾ while $\log(\sigma/\text{S cm}^{-1}) = -7.0$ for LT-LiBH_4 ⁵⁾ at 333 K), into a bulk-type all-solid-state TiS_2/Li battery. It was hypothesized that this would lower the battery operating temperature below 393 K.³³⁾ In addition, this material was expected to have higher oxidative stability than LiBH_4 .¹⁸⁾ The battery performance is discussed based on the electrochemical and thermal properties of the $\text{Li}_2\text{B}_{12}\text{H}_{12}$ electrolyte.

2. Experimental Procedure

2.1 Preparation of $\text{Li}_2\text{B}_{12}\text{H}_{12}$

In this study, we synthesized $\text{Li}_2\text{B}_{12}\text{H}_{12}$ via the solid-state reaction of LiBH_4 ($\geq 95\%$, Sigma-Aldrich) and $\text{B}_{10}\text{H}_{14}$ (Reagent grade, Wako Pure Chemical Industries, Ltd.) as reported previously.^{32,34)} The reagents were weighed in a stoichiometric ratio and mixed using an agate mortar in an agate pestle. The powder mixture was further mixed by planetary ball milling at 400 rpm for 5 h. The resultant powder was pressed into pellet, transferred into a molybdenum crucible, and annealed at 473 K for 15 h in an Ar environment. The product prepared by the solid-state reaction route is labeled as $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$. We also examined a commercially available $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 4\text{H}_2\text{O}$ powder (Katchem) was investigated. The powder was dehydrated by annealing at 498 K for 20 h in vacuum. The resultant powder is labeled as $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$. To identify the crystal structure of both $\text{Li}_2\text{B}_{12}\text{H}_{12}$ samples, powder X-ray diffraction (XRD, X'Pert Pro, PANalytical) measurements were carried out at room temperature.

2.2 Electrochemical measurements

The ionic conductivities of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ were investigated using a lithium-symmetric cell configuration and a two-probe ac technique. The powders were placed into an 8 mm die, and pressed at 240 MPa at room temperature. Lithium discs were

cut from a lithium foil (Honjo Metal Co., Ltd.) and placed onto both surfaces of the electrolyte compact. The lithium-symmetric cells were placed in a stainless steel electrochemical cell with an 8 mm diameter Teflon guide, as schematically illustrated elsewhere.¹⁷⁾ The assembled cell was connected to a Chemical Impedance Meter (3532-80, Hioki Corp.). The ac impedance measurements were carried out in the temperature range from room temperature to 423 K with 10 K step intervals for two heating and cooling cycles. The input voltage perturbation and the frequency range were set to be 50–100 mV and 1 M–4 Hz, respectively. Cyclic voltammograms were obtained at a scan rate of 5 mV s^{-1} for asymmetric stainless steel (SS) | $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ | Li cells at 393 K and 333 K in the voltage range from -0.1 to 5 V. DC relaxation curves were collected for lithium symmetric cells using the $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ electrolyte under a constant voltage of 20 mV at 393 K and 333 K for 1 h using an electroanalytical system (1470E, Solartron Analytical).

2.3 Battery assembly and testing

Commercially available TiS_2 powder (99.9%, Sigma-Aldrich) was used as the positive electrode active material. The TiS_2 powder and $\text{Li}_2\text{B}_{12}\text{H}_{12}$ were mixed in a 2:3 mass ratio using an agate pestle in an agate mortar. The resultant mixtures were then used as the composite positive electrode. At first, 25 mg of the $\text{Li}_2\text{B}_{12}\text{H}_{12}$ powder was placed in an 8 mm die, and then pressed at 60 MPa at room temperature. Subsequently, 6 mg of the composite positive electrode powder was transferred onto the pressed electrolyte still present in the die, and uniaxially pressed at 240 MPa to obtain a single pellet comprised of the composite positive electrode and the electrolyte layers.

The microstructures and element distributions of the bilayer compact comprised of the composite positive electrode and electrolyte layers were recorded using a scanning electron microscope (SEM, JSM-6009, JEOL Ltd.) and an energy dispersive X-ray spectrometer (EDS, EX-54175JMH, JEOL Ltd.), respectively. A lithium disc with an 8 mm diameter was cut from a lithium foil and used as the negative electrode. The lithium disc was placed opposite the positive electrode layer. The assembled bulk-type all-solid-state TiS_2/Li battery was placed in a stainless steel electrochemical cell, as in the case of the conductivity measurements described above. Battery tests were carried out at 393 K and 0.2 C, and 333 K and 0.05 C in the voltage range of 1.6–2.7 V using an electrochemical battery test system (580 Battery Test System, Scribner Associates, Inc.). The values 0.2 C and 0.05 C correspond to the current densities of $228 \mu\text{A cm}^{-2}$ and $57 \mu\text{A cm}^{-2}$, respectively. The batteries were held at the measurement temperatures for 2 h to observe the open circuit potential (OCP) prior to the battery test. An extra battery was assembled for the OCP measurement at 373 K to systematically observe the OCP values as a function of temperature.

In order to compare the battery performances, we assembled a battery consisting of the $\text{TiS}_2/\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$ positive electrode, the LiBH_4 electrolyte, and the Li negative electrode, and 20 mg of the LiBH_4 powder was used for the battery assembly. The procedure was the same as the battery assembly described above.

The reactivity of TiS_2 with electrolytes such as LiBH_4 ,

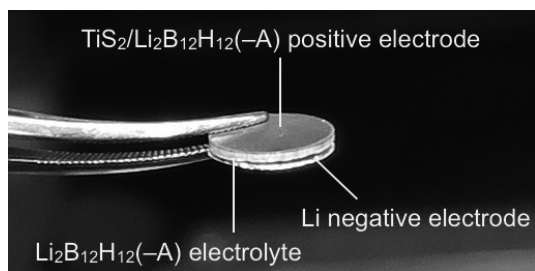


Fig. 2 A photograph of the bulk-type all-solid-state battery, $\text{TiS}_2/\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A}) | \text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A}) | \text{Li}$. A part of the negative electrode was delaminated to clearly show the battery configuration.

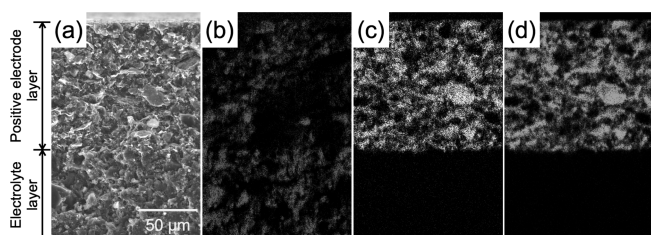


Fig. 3 Fractured surface image and element distributions of the bilayer compact consisting of the composite positive electrode, $\text{TiS}_2/\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$, and the electrolyte, $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$. (a) SEM image. Distributions of (b) B, (c) Ti, and (d) S.

$\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$, and $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$ was investigated by means of the powder XRD measurements at room temperature. TiS_2 and the electrolytes were mixed in a 2:3 mass ratio (the same mixing ratio as for the composite positive electrode for the battery assembly), pressed into a pellet and annealed at 393 K for 2 h. The resultant compacts were well ground and again annealed 10 times or until the solid-state reaction was likely finished.

3. Results and Discussions

3.1 Crystal structure of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ and battery assembly

Peaks in the XRD patterns obtained for both $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ and $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$ could be assigned to the cubic phase of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ with the space group of $Pa-3$.³⁵⁾ The lattice parameter, a , was 0.9535(3) nm for $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$, and 0.956009(9) nm for $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$. These values were similar to the literature data in which $a = 0.95771(2)$ nm.³⁵⁾

Regardless of the preparation method, $\text{Li}_2\text{B}_{12}\text{H}_{12}$ allowed for the assembly of the bulk-type all-solid-state batteries merely by uniaxial pressing at room temperature, i.e., cold-pressing similar to that used for LiBH_4 , because of their high deformability,^{7,16–20)} as a photograph shown in Fig. 2. Microstructures observed by SEM and element distributions evaluated by EDS, shown in Fig. 3, suggest that the electrolyte layer is fully densified, and robust interfaces are introduced between TiS_2 and $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$, and between the positive electrode and the electrolyte layers. This microstructure is ideal to reduce the contact resistance at the hetero-interfaces.³⁶⁾

3.2 Battery test

Repeated operations of the bulk-type all-solid-state $\text{TiS}_2/$

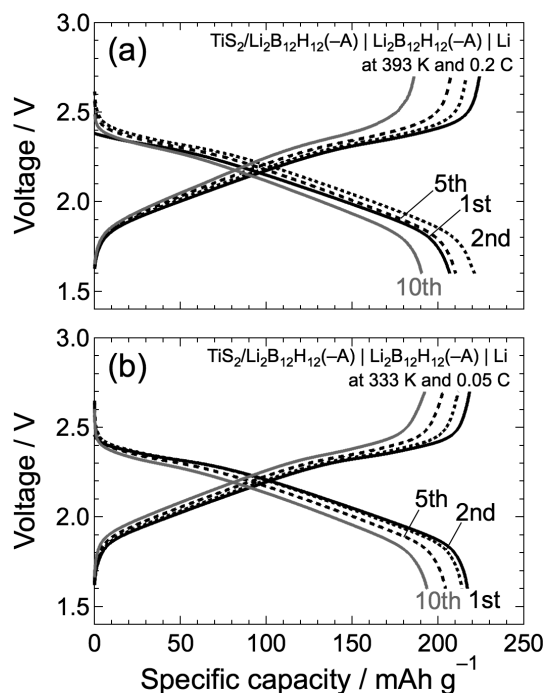
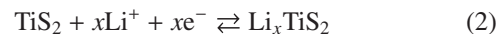


Fig. 4 Discharge-charge profiles of the bulk-type all-solid-state $\text{TiS}_2/\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A}) | \text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A}) | \text{Li}$ battery operated at (a) 393 K and 0.2 C ($228 \mu\text{A cm}^{-2}$), and (b) 333 K and 0.05 C ($57 \mu\text{A cm}^{-2}$).

Li batteries were successfully demonstrated at 393 K and 333 K when the $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ electrolyte was used, as the discharge-charge profiles shown in Figs. 4. The electrochemical intercalation and deintercalation of lithium-ions into TiS_2 and from Li_xTiS_2 , respectively, proceed as:^{37,38)}



The theoretical capacity is 239 mAh g^{-1} when x in Li_xTiS_2 varies from 0 to 1.

Our battery exhibited an initial discharge capacity of 207 mAh g^{-1} when operated at 393 K and 0.2 C. A high discharge capacity of 190 mAh g^{-1} was maintained at the tenth discharge. The high discharge capacity of the battery was retained even though the battery operating temperature were reduced to 333 K, and at 0.05 C. At this temperature, the initial discharge capacity was 217 mAh g^{-1} . The tenth discharge capacity remained as high as 193 mAh g^{-1} . Regardless of the battery operating temperature, our battery realized a high coulombic efficiency, which is defined by the ratio of the discharge capacity to the charge capacity, at the initial cycle as is expected for this electrolyte¹⁸⁾ and summarized in Table 1. The stability of this electrolyte will be discussed later. The battery test results suggest that both the high lithium ionic conductivity and the high oxidative stability of $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ ensured the repeated operation of the bulk-type all-solid-state battery with high coulombic efficiency.

On the other hand, repeated operation of the bulk-type all-solid-state battery that used the $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$ electrolyte was not possible, as may be seen from the discharge-charge profiles shown in Fig. 5 (a). This was in spite of a C-rate one order of magnitude smaller than the that used for the TiS_2/Li battery incorporating the $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ electrolyte at 393 K. Although further investigation is needed to fully understand

Table 1 Summary of open circuit potential (OCP) (vs. Li/Li⁺) just before the battery test, and the coulombic efficiency (defined by the ratio of charge capacity to discharge capacity) at the initial cycle of the bulk-type all-solid-state TiS₂/Li battery using the Li₂B₁₂H₁₂(-A) electrolyte. The OCPs of Li_xTiS₂ are 2.452 V and 1.664 V (vs. Li/Li⁺) when $x \sim 0$ and $x \sim 1$, respectively, at 393 K.¹⁸⁾

Battery operating temperature/K	OCP (vs. Li/Li ⁺)/V	Coulombic efficiency at the initial cycle/%
393	2.386	92
373	2.409	94
333	2.469	99

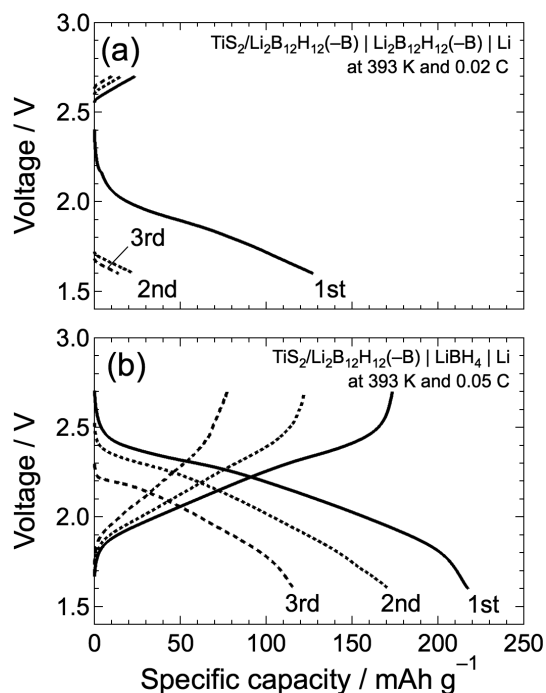


Fig. 5 Discharge-charge profiles of the bulk-type all-solid-state batteries: (a) TiS₂/Li₂B₁₂H₁₂(-B) | Li₂B₁₂H₁₂(-B) | Li operated at 393 K and 0.02 C (23 $\mu\text{A cm}^{-2}$), and (b) TiS₂/Li₂B₁₂H₁₂(-B) | LiBH₄ | Li operated at 393 K and 0.05 C (57 $\mu\text{A cm}^{-2}$).

this reason, large internal resistance is a possible factor that interferes the battery operation; thus, we evaluated the ionic conductivities of Li₂B₁₂H₁₂(-A) and Li₂B₁₂H₁₂(-B), as shown in Fig. 1. While Li₂B₁₂H₁₂(-A) possessed a lithium ionic conductivity of $\log(\sigma/\text{S cm}^{-1}) = -2.6$, Li₂B₁₂H₁₂(-B) had only $\log(\sigma/\text{S cm}^{-1}) = -4.4$ at 393 K. The high lithium ionic conductivity of $\log(\sigma/\text{S cm}^{-1}) = -3.5$ at 333 K for Li₂B₁₂H₁₂(-A) is thought to allow for battery operation at this temperature.

To assess the impact of lithium ionic conductivity on battery performance, the performance of the bulk-type all-solid-state TiS₂/Li battery that uses LiBH₄ as the electrolyte layer was evaluated, and the results are shown in Fig. 5 (b). The lithium ionic conductivity of LiBH₄ has been reported as $\log(\sigma/\text{S cm}^{-1}) = -2.7$.⁵⁾ The battery performance was improved compared with the battery comprised of only the Li₂B₁₂H₁₂(-B) electrolyte. The TiS₂/Li battery that uses LiBH₄ exhibited a high initial discharge capacity of 217 mAh g⁻¹, but the second and the third discharge capaci-

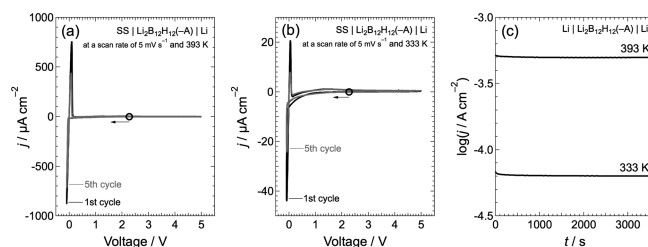


Fig. 6 Cyclic voltammograms of the asymmetric SS | Li₂B₁₂H₁₂(-A) | Li cells at a scan rate of 5 mV s⁻¹: (a) 393 K and (b) 333 K. (c) DC relaxation curves for the symmetric Li | Li₂B₁₂H₁₂(-A) | Li cell under 20 mV at 393 K and 333 K.

ties degraded and were 171 mAh g⁻¹ and 116 mAh g⁻¹, respectively. The lithium ionic conductivity of LiBH₄ is higher than that of Li₂B₁₂H₁₂(-B) at 393 K, and the lithium ionic conductivity of the latter compound in the composite positive electrode is considered insufficient. Hence, this limits the performance of the bulk-type all-solid-state TiS₂/Li battery.

3.3 Stability of Li₂B₁₂H₁₂

In the cyclic voltammograms obtained for the asymmetric SS | Li₂B₁₂H₁₂(-A) | Li cells, reversible reductive and oxidative currents were observed at around 0 V owing to lithium dissolution and precipitation at 393 K and 333 K, as shown in Fig. 6 (a) and (b). In addition, when dc voltage was applied to the lithium symmetric cells, the current rapidly relaxed and steady-state currents were observed, as displayed in Fig. 6 (c). These results suggest that the Li₂B₁₂H₁₂(-A) electrolyte forms a stable interface with the lithium electrode, as illustrated by the stable battery operation demonstrated for the bulk-type all-solid-state TiS₂/Li battery in Fig. 4 (a) and (b).

In the high voltage range, no irreversible oxidative wave was observed at the interface comprised of the Li₂B₁₂H₁₂(-A) and the SS electrode. The OCP of the bulk-type all-solid-state TiS₂/Li battery using the Li₂B₁₂H₁₂(-A) electrolyte was 2.386 V (vs. Li/Li⁺) although a value of 2.452 V is expected for Li_xTiS₂ ($x \sim 0$) at 393 K.¹⁸⁾ In addition, the coulombic efficiency at the initial cycle deviated from 100% at the higher battery operating temperatures. These results suggest that the intercalation of Li into TiS₂ via the solid-state reaction between TiS₂ and Li₂B₁₂H₁₂(-A) took place before the battery test in the composite positive electrode at the battery operating temperature. Since the interface stability is determined by the combination of the electrolyte and electrode materials, we decided to investigate the oxidative stability of Li₂B₁₂H₁₂ by investigating the reactivity with TiS₂.

To understand the reactivity between TiS₂ and Li₂B₁₂H₁₂(-A), we carried out the XRD measurements on the mixture obtained after repeated solid-state reaction cycle at 393 K. For comparison, we carried out the identical solid-state reactions and subsequent XRD measurements for the mixtures comprised of TiS₂, and LiBH₄ or Li₂B₁₂H₁₂(-B). The diffraction peaks resulting from TiS₂ shifted towards lower angles for the TiS₂/LiBH₄, and TiS₂/Li₂B₁₂H₁₂(-A) mixtures following annealing at 393 K for 2 h, as may be seen from the XRD patterns shown in Fig. 7 (a) and (b). These results suggest that a lattice expansion of TiS₂ took place in these mixtures. The lattice expansion of TiS₂ in the

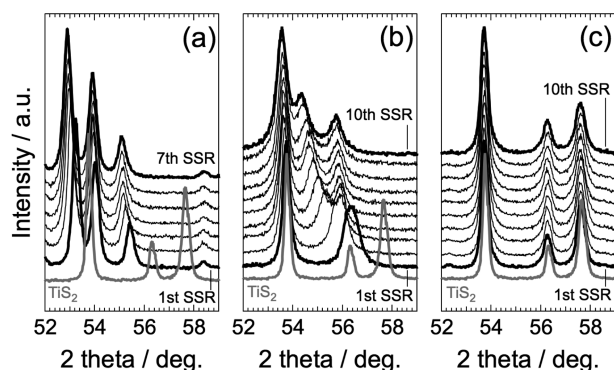


Fig. 7 Powder XRD patterns of the mixtures of (a) TiS_2 and LiBH_4 , (b) TiS_2 and $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$, and (c) $\text{TiS}_2/\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$. The solid-state reaction was carried out at 393 K for 2 h in an Ar environment, and it was repeated ten times for until the solid-state reaction was likely finished.

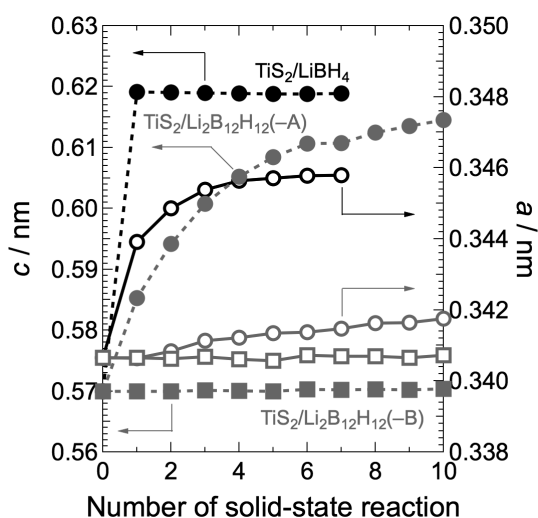


Fig. 8 Lattice parameters of Li_xTiS_2 in the mixtures of TiS_2 and LiBH_4 , TiS_2 and $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$, and TiS_2 and $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$ as a function of the number of solid-state reactions.

$\text{TiS}_2/\text{LiBH}_4$ mixture was more obvious than that in the $\text{TiS}_2/\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ mixture. On the other hand, as shown in Fig. 7 (c), no peak shifts were observed for the mixtures of $\text{TiS}_2/\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$ despite the fact that the solid-state reaction cycle was repeated 10 times. From the XRD patterns of the mixture, we evaluated the lattice parameters of Li_xTiS_2 by assuming that the lattice expansion of TiS_2 was caused only by the intercalation of lithium into TiS_2 , and the results are depicted in Fig. 8.

Before the annealing, our TiS_2 had a hexagonal structure with a space group of $P-3m1$, and the lattice parameters were $a = 0.34061(2)$ nm and $c = 0.56986(7)$ nm.¹⁸⁾ These agree with the literature data of $a = 0.34079(3)$ nm and $c = 0.56989(6)$ nm.³⁹⁾ For the $\text{TiS}_2/\text{LiBH}_4$ mixture, the lattice parameters were $a = 0.34580(2)$ nm and $c = 0.61885(4)$ nm after the seventh solid-state reaction. These parameters agree with the literature data for LiTiS_2 , i.e., $a = 0.34590(3)$ nm and $c = 0.61879(6)$ nm.³⁸⁾ This means that the solid-state reaction cycle between TiS_2 and LiBH_4 to form LiTiS_2 was finished by the seventh solid-state reaction. Since Li_xTiS_2 has a redox potential of 1.664 V (vs. Li/Li^+) when $x \sim 1$,¹⁸⁾ the decomposition potential of LiBH_4 is lower than this voltage at 393 K.

For the $\text{TiS}_2/\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ mixture, the lattice parameters for Li_xTiS_2 were $a = 0.34176(5)$ nm and $c = 6.145(1)$ nm. From the literature data describing the lattice parameters as a function of x in Li_xTiS_2 ,⁴⁰⁾ we could estimate x in Li_xTiS_2 to be $x = 0.3\text{--}0.4$. This result suggests that $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ is less reactive than LiBH_4 . Thus, our battery realized a high OCP before the battery test and a high coulombic efficiency over the battery operation, as shown in Fig. 4. It should be noted here that the reaction between TiS_2 and $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ was unlikely to be finished after ten solid-state reaction cycles. Thus, further investigations are required to fully address the thermodynamics of $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$.

^1H MAS-NMR spectra of $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ inferred the existence of a non-negligible concentration of the $[\text{B}_{10}\text{H}_{10}]^{2-}$ cluster anion.³²⁾ As in this previous report, we also observed small concentrations of $[\text{BH}_4]^-$ and $[\text{B}_{10}\text{H}_{10}]^{2-}$ in the ^1H MAS-NMR spectra of the $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$ electrolyte synthesized in this study. In addition, the differential scanning calorimetry (DSC) profile at elevated temperature exhibited a sharp endothermic peak appeared at 630 K for $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$, accompanied by a phase transition.⁴¹⁾ Such a clear endothermic peak was not observed for the $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$. Gas chromatograph (G.C.) profiles at elevated temperatures show that $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{B})$ underwent a one-step pyrolysis reaction at temperatures around 750–850 K. On the other hand, the situation seems to be more complicated for $\text{Li}_2\text{B}_{12}\text{H}_{12}(-\text{A})$. This material had more than two pyrolysis steps, appearing at 570–750 K and higher than 770 K (^1H MAS-NMR spectra, DSC and GC profiles are not shown). These results suggest that the controls of ionic conduction and thermodynamic properties of the complex hydrides containing cluster anion would be possible by utilizing crystal modifications, which are different from the conventional element replacement method, typically replacement of the complex anion by halides.³¹⁾ In addition, we recently discovered that when $\text{Na}_2\text{B}_{12}\text{H}_{12}$ powder is aggressively ball-milled, the crystals become nanometer scaled, which results in the disappearance of the conductivity jump accompanied by the phase transition.²⁹⁾ As a result, ball-milled $\text{Na}_2\text{B}_{12}\text{H}_{12}$ possessed high sodium ionic conductivity of $\log(\sigma/\text{S cm}^{-1}) = -3.8$ at room temperature, as shown in Fig. 1. Our future studies will focus on how the chemical composition and microstructure of the complex hydrides with cluster anions impact on the ionic conductivity and phase stabilities.

4. Conclusion

In this study, we incorporated the $\text{Li}_2\text{B}_{12}\text{H}_{12}$ electrolyte into a bulk-type all-solid-state TiS_2/Li battery. Repeated operations were successfully demonstrated for at least ten discharge–charge cycles, retaining high discharge capacities, at 393 K and 333 K. At these temperatures, this material has a high lithium ionic conductivity of $\log(\sigma/\text{S cm}^{-1}) = -2.6$ at 393 K, and -3.5 at 333 K. Compared with the bulk-type all-solid-state TiS_2/Li battery that uses the LiBH_4 electrolyte, our battery maintained high initial discharge capacity and high coulombic efficiency during the repeated discharge–charge cycles. These are owing to the superior oxidative stability of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ compared with LiBH_4 . In future studies, we will investigate the detailed chemical composition, micro-

structure and phase stability of this class of materials with a view toward the development of high-performance novel solid-state electrolytes for use in batteries.

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REFERENCES

- 1) L. Schlapbach and A. Züttel: *Nature* **414** (2001) 353–358.
- 2) S. Orimo, Y. Nakamori, J.R. Eliseo, A. Züttel and C.M. Jensen: *Chem. Rev.* **107** (2007) 4111–4132.
- 3) W. Zaïdi, J.-P. Bonnet, J. Zhang, F. Cuevas, M. Katroche, S. Couillaud, J.-L. Bobet, M.T. Sougrati, J.-C. Jumas and L. Aymard: *Int. J. Hydrogen Energ.* **38** (2013) 4798–4808.
- 4) J. Zhang, W. Zaïdi, V. Paul-Boncour, K. Provost, A. Michalowicz, F. Cuevas, M. Latroche, S. Belin, J.-P. Bonnet and L. Aymard: *J. Mater. Chem. A* **1** (2013) 4706–4717.
- 5) M. Matsuo, Y. Nakamori, S. Orimo, H. Maekawa and H. Takamura: *Appl. Phys. Lett.* **91** (2007) 224103.
- 6) M. Matsuo and S. Orimo: *Adv. Energy Mater.* **1** (2011) 161–172.
- 7) A. Unemoto, M. Matsuo and S. Orimo: *Adv. Funct. Mater.* **24** (2014) 2267–2279.
- 8) J.-P. Soulie, G. Renaudin, R. Cerny and K. Yvon: *J. Alloy. Compd.* **346** (2002) 200–205.
- 9) A. Züttel, S. Rentsch, P. Fischer, P. Wenger, P. Sudan, P. Mauron and C. Emmenegger: *J. Alloy. Compd.* **356–357** (2003) 515–520.
- 10) M.P. Pitt, M. Paskevicius, D.H. Brown, D.A. Sheppard and C.E. Buckley: *J. Am. Chem. Soc.* **135** (2013) 6930–6941.
- 11) S. Orimo, Y. Nakamori, N. Ohba, K. Miwa, S. Towata, A. Züttel and M. Aoki: *Appl. Phys. Lett.* **89** (2006) 021920.
- 12) N. Ohba, K. Miwa, M. Aoki, T. Noritake, S. Towata, Y. Nakamori, S. Orimo and A. Züttel: *Phys. Rev. B* **74** (2006) 075110.
- 13) T. Ikeshoji, E. Tsuchida, K. Ikeda, M. Matsuo, H.-W. Li, Y. Kawazoe and S. Orimo: *Appl. Phys. Lett.* **95** (2009) 221901.
- 14) T. Ikeshoji, E. Tsuchida, T. Morishita, K. Ikeda, M. Matso, Y. Kawazoe and S. Orimo: *Phys. Rev. B* **83** (2011) 144301.
- 15) J.S.G. Myrdal, B. Blanchard, D. Sveinbjörnsson and T. Vegge: *J. Phys. Chem. C* **117** (2013) 9084–9091.
- 16) A. Unemoto, H. Wu, T.J. Udovic, M. Matsuo, T. Ikeshoji and S. Orimo: *Chem. Commun. (Camb.)* **52** (2016) 564–566.
- 17) A. Unemoto, S. Yasaku, G. Nogami, M. Tazawa, M. Taniguchi, M. Matsuo, T. Ikeshoji and S. Orimo: *Appl. Phys. Lett.* **105** (2014) 083901.
- 18) A. Unemoto, T. Ikeshoji, S. Yasaku, M. Matsuo, V. Stavila, T.J. Udovic and S. Orimo: *Chem. Mater.* **27** (2015) 5407–5416.
- 19) A. Unemoto, C. Chen, Z. Wang, M. Matsuo, T. Ikeshoji and S. Orimo: *Nanotechnology* **26** (2015) 254001.
- 20) K. Yoshida, S. Suzuki, J. Kawaji, A. Unemoto and S. Orimo: *Solid State Ionics* **285** (2016) 96–100.
- 21) A. Brazier, L. Dupont, L. Dantras-Laffont, N. Kuwata, J. Kawamura and J.-M. Tarascon: *Chem. Mater.* **20** (2008) 2352–2359.
- 22) A. Sakuda, A. Hayashi and M. Tatsumisago: *Chem. Mater.* **22** (2010) 949–956.
- 23) N. Ohta, K. Takada, L. Zhang, R. Ma, M. Osada and T. Sasaki: *Adv. Mater.* **18** (2006) 2226–2229.
- 24) J. Haruyama, K. Sodeyama, L. Han, K. Takada and Y. Tateyama: *Chem. Mater.* **26** (2014) 4248–4255.
- 25) K. Takada: *Acta Mater.* **61** (2013) 759–770.
- 26) W.S. Tang, A. Unemoto, W. Zhou, V. Stavila, M. Matsuo, H. Wu, S. Orimo and T.J. Udovic: *Energy Environ. Sci.* **8** (2015) 3637–3645.
- 27) W.S. Tang, M. Matsuo, H. Wu, V. Stavila, W. Zhou, A.A. Talin, A.V. Soloninin, R.V. Skoryunov, O.A. Babanova, A.V. Skripov, A. Unemoto, S. Orimo and T.J. Udovic: *Adv. Energy Mater.* **6** (2016) 1502237.
- 28) T.J. Udovic, M. Matsuo, A. Unemoto, N. Verdal, V. Stavila, A.V. Skripov, J.J. Rush, H. Takamura and S. Orimo: *Chem. Commun. (Camb.)* **50** (2014) 3750–3752.
- 29) W.S. Tang, M. Matsuo, H. Wu, V. Stavila, A. Unemoto, S. Orimo and T.J. Udovic: *Energ. Storage Mater.* **4** (2016) 79–83.
- 30) T.J. Udovic, M. Matsuo, W.S. Tang, H. Wu, V. Stavila, A.V. Soloninin, R.V. Skoryunov, O.A. Babanova, A.V. Skripov, J.J. Rush, A. Unemoto, H. Takamura and S. Orimo: *Adv. Mater.* **26** (2014) 7622–7626.
- 31) H. Maekawa, M. Matsuo, H. Takamura, M. Ando, Y. Noda, T. Karahashi and S. Orimo: *J. Am. Chem. Soc.* **131** (2009) 894–895.
- 32) L. He, H.-W. Li, H. Nakajima, N. Tumanov, Y. Filinchuk, S.-J. Hwang, M. Sharma, H. Hagemann and E. Akiba: *Chem. Mater.* **27** (2015) 5483–5486.
- 33) J.A. Teprovich, Jr., H. Colon-Mercado, A.L. Washington, II, P.A. Ward, S. Greenway, D.M. Missimer, H. Hartman, J. Velten, J.H. Christian and R. Zidan: *J. Mater. Chem. A* **3** (2015) 22853–22859.
- 34) L. He, H.-W. Li, S.-J. Hwang and E. Akiba: *J. Phys. Chem. C* **118** (2014) 6084–6089.
- 35) J.-H. Her, M. Yousufuddin, W. Zhou, S.S. Jalisatgi, J.G. Kulleck, J.A. Zan, S.-J. Hwang, R.C. Bowman, Jr. and T.J. Udovic: *Inorg. Chem.* **47** (2008) 9757–9759.
- 36) A. Sakuda, A. Hayashi and M. Tatsumisago: *Sci. Rep.* **3** (2013) 2261.
- 37) M.S. Whittingham: *Science* **192** (1976) 1126–1127.
- 38) M.S. Whittingham: *Prog. Solid State Chem.* **12** (1978) 41–99.
- 39) J. Dahn, W. McKinnon, R.R. Haering, W. Buyers and B. Powerll: *Can. J. Phys.* **58** (1980) 207–213.
- 40) J. Dahn and R.R. Haering: *Solid State Commun.* **40** (1981) 245–248.
- 41) M. Paskevicius, M.P. Pitt, D.H. Brown, D.A. Sheppard, S. Chumphongphan and C.E. Buckley: *Phys. Chem. Chem. Phys.* **15** (2013) 15825.